

National Computer Education Accreditation Council NCEAC

NCEAC.FORM.001-D

COURSE DESCRIPTION FORM

INSTITUTION FAST School of Computing, National University of Computer and Emerging Sciences, Islamabad

BS-CS Spring 2025

**PROGRAM (S) TO BE
EVALUATED**

Course Description

Course Code	CS4032																		
Course Title	Web Programming																		
Credit Hours	3																		
Prerequisites by Course(s) and Topics	NA																		
Grading Policy	Absolute Grading																		
Policy about missed assessment items in the course	Retake of missed assessment items (other than sessional/ final exam) will not be held. Student who misses an assessment item (other than sessional / final exam) is awarded zero marks in that assessment item i.e., late submission will not be accepted. For missed sessional/ final exam, exam retake/ pretake application along with necessary evidence are required to be submitted to the department secretary. The examination assessment and retake committee decide the exam retake/ pretake cases.																		
Course Plagiarism Policy	Plagiarism in project or midterm/ final exam may result in F grade in the course. Plagiarism in an assignment will result in zero marks in the whole assignments category.																		
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	70% Theory + 30% In class Lab Assessment items of Theory and Lab Part <table><tr><th>Assessment Item</th><th>Number</th><th>Weight (%)</th></tr><tr><td>Assignments</td><td>>=3</td><td>20</td></tr><tr><td>Quizzes/Tasks</td><td>>=5</td><td>15</td></tr><tr><td>Sessional</td><td>2</td><td>15</td></tr><tr><td>Project</td><td>1</td><td>15</td></tr><tr><td>Final Exam</td><td>1</td><td>35</td></tr></table>	Assessment Item	Number	Weight (%)	Assignments	>=3	20	Quizzes/Tasks	>=5	15	Sessional	2	15	Project	1	15	Final Exam	1	35
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Course Instructors	Dr. Tajwar Mehmood																		
Lab Instructors (if any)	NA																		

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Course Coordinator	Dr. Tajwar Mehmood																										
URL (if any)	Google Classroom Class code: 3wylxd3f																										
Current Catalog Description	Web Development Technologies, Frontend & backend technologies, HTML/HTML5, CSS/CSS3, JavaScript, TypeScript, NodeJS, Angular, ReactJS, ExpressJS, MongoDB, MEAN Stack, MERN Stack, Serverless Stack Development and best practices in web programming.																										
Textbook (or Laboratory Manual for Laboratory Courses)	Web Application Architecture Principles, protocols and practices by Leon Shklar and Richard Rosen The Missing Link: An Introduction to Web Development and Programming by Michael Mendez																										
Reference Material	Learning JavaScript , 3 rd Edition by Todd Brown Full Stack JavaScript Development with MEAN By Adam Bretz & Colin J. Ihrig Full-Stack React Projects , Second Edition by Shama Hoque																										
Course Learning Outcomes	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <th colspan="3" style="background-color: #d3d3d3;">A. Course Learning Outcomes (CLOs)</th></tr> <tr> <td colspan="3">After completion of the course, the students shall be able to:</td></tr> <tr> <td colspan="3"> 1. Develop and design web applications using modern web development frameworks 2. Learn how to use different web frameworks to create a complete industry-oriented project using MEAN and MERN Stack 3. Able to design, develop, and deploy full stack web applications 4. Work in a team to complete enterprise projects and professional ethics and responsibilities. </td></tr> <tr> <th colspan="3" style="background-color: #d3d3d3;">B. Program Learning Outcomes</th></tr> <tr> <td colspan="3">For each attribute below, indicate whether this attribute is covered in this course or not. Leave the cell blank if the enablement is little or non-existent.</td></tr> <tr> <td style="width: 20%;">1. Computing Knowledge:</td><td style="width: 60%;">Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution of complex computing problems.</td><td style="width: 20%; text-align: center;">✓</td></tr> <tr> <td>2. Problem Analysis:</td><td>Identify, formulate, research literature, and analyze complex computing problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, and computing sciences.</td><td style="text-align: center;">✓</td></tr> <tr> <td>3. Design/Develop Solutions:</td><td>Design solutions for complex computing problems and design systems, components, and processes that meet specified needs with appropriate</td><td style="text-align: center;">✓</td></tr> </table>			A. Course Learning Outcomes (CLOs)			After completion of the course, the students shall be able to:			1. Develop and design web applications using modern web development frameworks 2. Learn how to use different web frameworks to create a complete industry-oriented project using MEAN and MERN Stack 3. Able to design, develop, and deploy full stack web applications 4. Work in a team to complete enterprise projects and professional ethics and responsibilities.			B. Program Learning Outcomes			For each attribute below, indicate whether this attribute is covered in this course or not. Leave the cell blank if the enablement is little or non-existent.			1. Computing Knowledge:	Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution of complex computing problems.	✓	2. Problem Analysis:	Identify, formulate, research literature, and analyze complex computing problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, and computing sciences.	✓	3. Design/Develop Solutions:	Design solutions for complex computing problems and design systems, components, and processes that meet specified needs with appropriate	✓
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		consideration for public health and safety, cultural, societal, and environmental considerations.	
	4. Investigation; Experimentation:	Conduct investigation of complex computing problems using research based knowledge and research based methods	✓
	5. Modern Tool Usage:	Create, select, and apply appropriate techniques, resources and modern computing tools, including prediction and modelling for complex computing problems.	✓
	6. Society Responsibility:	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues relevant to context of complex computing problems.	✓
	7. Environment and Sustainability:	Understand and evaluate sustainability and impact of professional computing work in the solution of complex computing problems.	✓
	8. Ethics:	Apply ethical principles and commit to professional ethics and responsibilities and norms of computing practice.	✓
	9. Individual and Team Work:	Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.	✓
	10. Communication:	Communicate effectively on complex computing activities with the computing community and with society at large.	✓
	11. Project Management and Finance:	Demonstrate knowledge and understanding of management principles and economic decision making and apply these to one's own work as a member or a team.	✓
	12. Life-long Learning:	Recognize the need for, and have the preparation and ability to engage in independent and	✓

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		life-long learning in the broadest context of technological changes.											
C. Mapping of CLOs on PLOs (CLO: Course Learning Outcome, PLOs: Program Learning Outcomes)													
		PLOs											
		1	2	3	4	5	6	7	8	9	10	11	12
C L O s	1	✓	✓	✓		✓							✓
	2	✓	✓	✓		✓							✓
	3	✓		✓									✓
	4	✓	✓	✓	✓	✓		✓				✓	✓
	5	✓		✓		✓							✓
	6	✓		✓			✓		✓	✓	✓	✓	
	7		✓		✓			✓				✓	✓

Topics Covered in the Course, with Number of Lectures on Each Topic (assume 15-week instruction and one-hour lectures)	Topics to be covered:			
	List of Topics	No. of Weeks	Contact Hours	CLO(s)
	Introduction to Web Development Front-end vs Back-end Development	1	3	1,2,7,12
	HTML, HTML5, CSS, CSS3, Bootstrap, tailwind	2	6	1,2,3,5,8
	JavaScript fundamentals , JQuery	1	3	1,2,3,5,8
	AJAX, JQUERY, FETCH, AXIOS	1	3	1,2,3,5,8
	State management and Data Bindings	1	3	1,3,4,12
	MongoDB	1	3	5,12,6
	Introduction to Node	1	3	1,2,3,5
	Introduction to Express	1	3	1,2,3,5
	Programming with Angular and Typescript	2	6	1,3,4,5,11,12
	Introduction to React Asynchronous JavaScript	2	6	1,2,3,5

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	Introduction to Vue	1	3	1,2,3,5
	Advance topics- Serverless , GraphQL	1	6	1,2,3,4, 5,12
	Deployment and Web Programming practices and Demos	1	3	1,6,7,9,10,11
	Total	16	48	
Laboratory Projects/Experiments Done in the Course	Introduction to HTML/HTML5, CSS/CSS3, JavaScript, TypeScript, State Management Techniques, Angular, Angular MVC/MVVM, NodeJS, MongoDB, ReactJS, Serverless Stack Development is managed with practical quizzes and project.			
Programming Assignments Done in the Course	Web Programming using HTML, CSS, JavaScript, TypeScript, Angular, Angular MVC/MVVM, NodeJS, ReactJS, Serverless Stack.			
Class Time Spent on (in credit hours, Hrs/Min)	Theory	Problem Analysis	Solution Design	Social and Ethical Issues
	20	25	50	5
Oral and Written Communications	Every student is required to submit at least <u> 1 </u> written reports of typically <u> 1 </u> pages and to make <u> 1 </u> oral presentation of typically <u> 5 </u> minutes' duration.			

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Lab/ Practical Component of the course

COURSE CONTENTS (Lab/ Practical):			
Weeks	Contents/Topics	**Courseware Events (MM/ IT Lab/Case Study/ Assignment/ Presentation etc.)	Comments (if any)
Week-01	Introduction to web technologies	Task0	
Week-02	Developing web pages using HTML and HTML5		
Week-03	Designing web pages using CSS and CSS3	Task 1	
Week-04	Application of Java Script for Web Application development	Assignment 1, Task2	
Week-05	State Management Techniques and Data binding	Task3	
Week-06	Developing database using MongoDB	Assignment 2	
Week-07	Introduction to NodeJS	Task4	
Week-08	Programming with ExpressJS and NodeJS	Task 5	
Week-09	Programming with Angular and TypeScript	Assignment 3	
Week-10	Programming with Angular and TypeScript	Task 6	
Week-11	Programming with ReactJS		
Week-12	Programming with ReactJS	Task 7	
Week-13	Hands on experience on MEAN and MERN Stack	Assignment 4, Project	
Week-14	NEXT JS	Task 8	
Week-15	Hands on practice on Serverless Stack Development	Assignment 5	

Practical/ Programming Work/ Tools: Visual Studio Code, Dreamweaver, Angular CLI, Node, MongoDB.